

CODE:	COURSE NAME:	CATEGORY	L	T	P	CREDIT
221EIT002	CLOUD COMPUTING AND SECURITY	PROGRAM ELECTIVE I	3	0	0	3

Preamble: The syllabus is designed with the view of preparing the students capable of understanding the security aspects of cloud computing. The students will get to know about the cloud architecture, vulnerabilities and threats occurring in various cloud systems and methods to secure the same.

Course Outcomes: Upon completion, the student will be able to

CO 1	Apply the basics of virtualization to implement virtual machines(Cognitive Level: Apply)
CO 2	Apply the fundamentals of cloud computing and its architecture for implementing cloud services.(Cognitive level: Apply)
CO 3	Deploy an application in cloud.(Cognitive level: Apply)
CO 4	Analyse the security issues and challenges faced in a given cloud environment and implement appropriate security measures(Cognitive level: Apply)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

- PO1:** An ability to independently carry out research/investigation and development work in engineering and allied streams
- PO2:** An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.
- PO3:** An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1							
CO 2							
CO 3							
CO 4							

Assessment Pattern	
Bloom's Category	End Semester Examination
Apply	100%
Analyse	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation: 40 marks

- i. Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- ii. Course based task / Seminar/ Data collection and interpretation : 15 marks
- iii. Test paper (1 number) : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.
There will be two parts; Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks

Total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Demonstrate how you will a Configure a VM instance using virtual box in your local machine and allocate CPU, memory and storage space as per a specified requirement.
2. Illustrate the steps to perform VM migration.
3. A degree of caution is advised when launching a server virtualization. Clarify this statement by stating the drawbacks of the same.

Course Outcome 2 (CO2):

1. With the help of examples, differentiate between SaaS, PaaS and IaaS?
2. To meet the requirements of a single organization, which one out of the private, public and hybrid cloud providers, is recommended? Why?
3. How will you classify massive systems? Explain.

Course Outcome 3 (CO3):

1. “OpenNebula and OpenStack serve different needs”. Justify this statement.
2. Demonstrate the challenges in designing and developing a cloud application, with the help of an example.
3. How is OpenStack used in a cloud environment?

Course Outcome 4 (CO4):

1. Given an e-healthcare application deployed in cloud. Identify the challenges faced and the security objectives to be achieved.
2. How is security provided to data at various stages in the context of the cloud?
3. How virtualization supports disaster recovery?

Model Question Paper:

QP CODE:

Reg No:

Name:

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR

221EIT002

CLOUD COMPUTING AND SECURITY

Max. Marks : 60

Duration: 2.5 hrs

PART A

Answer All Questions. Each Question Carries 5 Marks

1. Give the steps to install a virtual machine. (5)
 2. In which scenario, users can opt IaaS, PaaS and SaaS? Find suitable application for each. (5)
 3. How inter-cloud resource management is performed? (5)
 4. Among openNebula and openStack, which is suited best for enterprise cloud computing? (5)
 5. Why is it harder to establish security in the cloud? (5)
- (5x5=25)

Part B

(Answer any five questions. Each question carries 7 marks)

6. How is memory virtualization implemented? Provide necessary examples and diagrams wherever necessary. (7)
7. Illustrate the steps to perform live VM migration. (7)

8. Depict the cloud architecture. Explain the same with the help of an example application. (7)
9. Explain PaaS model for cloud computing. With a neat diagram, write about Google App Engine for PaaS applications. (7)
10. Consider a restaurant application on the cloud that has serverless apps connecting to a backend NoSQL database. There are two types of users in the application - manager (who manages the catalog) and customer (who views the catalog). There are separate serverless functions for the users. During development, you need to create a set of roles that can define what operations are permitted on the NoSQL database by the serverless functions. In addition, the entire application and the data should be secured. Propose solutions for the following. (7)
 - a) What permissions are required for the role assigned to manager function?
 - b) What permissions are required for the role assigned to customer function?
 - c) How can you ensure application security?
 - d) How can you ensure data security?
 - e) How can you create an auditable trail of modifications to the database?
11. Explain Security Architecture Design in cloud. (7)
12. Explain the life cycle of Secure software Development (SecSDLC). (7)

Syllabus

Module I:

Introduction to component virtualization: Basics of Virtualization - Types of Virtualization - Implementation Levels of Virtualization -Virtualization of CPU, Memory, I/O Devices – Virtual clusters and resource management- Desktop Virtualization – Server Virtualization- Storage Virtualization – Network Virtualization.

Module II: Introduction to Cloud Computing: System Models for Distributed and Cloud Computing - NIST Cloud Computing Reference Architecture - Features of Cloud Computing – Cloud Services – IaaS, PaaS, SaaS – Cloud service Providers – Public, Private and Hybrid Clouds.

Module III: Architectural Design of Compute and Storage Clouds – Layered Cloud Architecture Development – Design Challenges - Inter Cloud Resource Management – Resource Provisioning and Platform Deployment – Global Exchange of Cloud Resources.

Module IV: Parallel and Distributed Programming Paradigms – Map Reduce, Hadoop Library from Apache – Mapping Applications - Programming Support – EC2, Google App Engine, Amazon AWS - Cloud Software Environments - Eucalyptus, Open Nebula, OpenStack.

Security in the cloud: Overview – Cloud Security Challenges- Software-as-a-Service Security – Security Governance – Risk Management.

Module V:SecSDLC-Security Monitoring – Security Architecture Design – Data Security – Application Security. VM security. Cloud Security Architecture: Trusted cloud, Identity and access management, Access control in cloud, Autonomic security, protection, self-healing. Firewalls-Web application firewall.

Reference Books

1. Kai Hwang, Geoffrey C Fox, Jack G Dongarra, “Distributed and Cloud Computing, From Parallel Processing to the Internet of Things”, 1sted. ,Morgan Kaufmann Publishers.
2. John W.Rittinghouse and James F.Ransome, “Cloud Computing: Implementation, Management, and Security”, CRC Press, 2010.
3. Tim Mather, S. Kumaraswamy and S. Latif, “Cloud Security and Privacy: An EnterprisePerspective on Risks and Compliance”, O’Reilly Media, 2009.
4. Ronald L. Krutz Russell Dean Vines “Cloud Security: A Comprehensive Guide to SecureCloud Computing”, Wiley ,2010.

5. Toby Velte, Anthony Velte, Robert Elsenpeter, “Cloud Computing, A Practical Approach”, TMH, 2009.

6. George Reese, “Cloud Application Architectures: Building Applications and Infrastructure in the Cloud” O'Reilly, 2009.

7. James E. Smith, Ravi Nair, “Virtual Machines: Versatile Platforms for Systems and Processes”, Elsevier/Morgan Kaufmann, 2005.

8. Katarina Stanoevska-Slabeva, Thomas Wozniak, Santi Ristol, “Grid and Cloud Computing – A Business Perspective on Technology and Applications”, Springer, 2010.

Course Contents and Lecture Schedule

No	Topic	No. of Lectures (40)
1	Introduction to component virtualization:	8 Hours
1.1	Basics of Virtualization – Types of Virtualization	1 hour
1.2	Implementation Levels of Virtualization	1 hour
1.3	Virtualization of CPU, Memory, I/O Devices	1 hour
1.4	Virtual clusters and resource management	1 hour
1.5	Live VM migration steps	1 hour
1.6	Migration of memory, files and network resources	1 hour
1.7	Desktop Virtualization –Server Virtualization Storage Virtualization	1 hour
1.8	Network Virtualization	1 hour
2	Introduction to Cloud Computing	6 Hours

INFORMATION TECHNOLOGY-IT1

2.1	System Models for Distributed and Cloud Computing	1 hour
2.2	NIST Cloud Computing Reference Architecture	1 hour
2.3	Features of Cloud Computing	1 hour
2.4	Cloud Services – IaaS, PaaS, SaaS	1 hour
2.5	Cloud service Providers – Public	1 hour
2.6	Cloud service Providers – Private and Hybrid Clouds.	1 hour
3	Architectural Design of Compute and Storage Clouds	6 Hours
3.1	Architectural Design of Compute and Storage Clouds	1 hour
3.2	Layered Cloud Architecture Development	1 hour
3.3	Design Challenges	1 hour
3.4	Inter Cloud Resource Management	1 hour
3.5	Resource Provisioning and Platform Deployment	1 hour
3.6	Global Exchange of Cloud Resources.	1 hour
4	Paradigms and Environments	10 Hours
4.1	Parallel and Distributed Programming Paradigms – Map Reduce	1 hour
4.2	Twister and Iterative MapReduce	1 hour
4.3	Hadoop Library from Apache – Mapping Applications – Programming Support	1 hour
4.4	Google App Engine	1 hour

4.5	Amazon AWS	1 hour
4.6	Cloud Software Environments – Eucalyptus	1 hour
4.7	Open Nebula, OpenStack.	1 hour
4.8	Security in the cloud-overview	1 hour
4.9	Cloud security challenges	1 hour
4.10	Software-as-a-Service Security – Security Governance – Risk Management	1 hour
5	Cloud security	10 Hours
5.1	Secure Software Development Lifecycle	1 hour
5.2	Security Monitoring	1 hour
5.3	Security Architecture Design	1 hour
5.4	Data Security – Application Security. VM security	1 hour
5.5	Cloud Security Architecture: Trusted cloud,	1 hour
5.6	Identity and access management,	1 hour
5.7	Access control in cloud, RBAC	1 hour
5.8	Autonomic security, protection, self-healing.	1 hour
5.9	Firewalls	1 hour
5.10	Web application firewall.	1 hour

CODE	COURSE NAME	INFORMATION	IT	CH	NO	CREDITS
221EIT003	ETHICAL HACKING	PROGRAM ELECTIVE I	3	0	0	3

Preamble: The students are expected to understand the basic concepts of Ethical Hacking and Penetration testing.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Perform system security vulnerability testing
CO 2	Learn various hacking methods
CO 3	Perform system vulnerability exploit attacks
CO 4	Produce a security assessment report
CO 5	Recognize tools and methods to carry out penetration testing.

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1:An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2:An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3:An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor's program

PO4:An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards

PO5:An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tools to model, analyze and solve practical engineering problems.

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PO7:An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4 INFC	PO 5 INFORMATION	PO 6 TECHNOLOGY	PO 7-IT1
CO 1	✓		✓	✓	✓	✓	
CO 2	✓		✓	✓	✓	✓	
CO 3	✓		✓	✓	✓	✓	
CO 4	✓	✓	✓	✓	✓	✓	
CO 5	✓	✓	✓	✓	✓	✓	

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	80
Analyse	20
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation Pattern:

The evaluation shall only be based on application, analysis or design-based questions (for both internal and end-semester examinations).

Continuous Internal Evaluation: 40 marks

- Preparing a review article based on peer-reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- Course based task / Seminar/ Data collection and interpretation : 15 marks
- Test paper (1 number) : 10 marks

Test paper shall include a minimum of 80% of the syllabus.

Course-based task/test paper questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.

There will be two parts; Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum of one question from each module of which student should answer any five. Each question can carry 7 marks

The total duration of the examination will be 150 minutes.

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.....SEMESTER

Branch:

Time: 2½ Hours, Maximum marks: 60

PART A (Answer all questions. Each question carries 5 marks)

1. Explain the terms Threat, Vulnerability, and Risk.
2. What are the different methods for managing insider threats?
3. What do you mean by 'Vulnerability Scanner'? Draw the scanner and briefly explain
4. Explain in detail computer fraud classification according to the data processing model?
5. Is there any method to demonstrate whether network infrastructure is adequate or not? Justify your answer.

PART B (Answer any five questions. Each question carries 7 marks)

6. Being an expert in hacking, you are asked to analyze the free, open-source software being used to run FTP services on a server. You noticed that there is an excessive number of `gets()` and `get()` on the source code. These C++ functions do not check bounds. What kind of attack is this program susceptible to? Comment on your recommendation to prevent this attack.
7. Explain the type of attack faced by the stack data structure. Also, explain the uses of two pointers used in a stack.
8. Illustrate how SQL injection attacks are done by attackers.
9. Discuss some architecture strategies for fraud prevention with respect to computer network-enabled system
10. Explain IT infrastructure.
11. An attacker uses a distributed group of computers to shut down a single machine or network making it inaccessible to its intended users? Explain the attack with the help of a neat diagram. What are the ways to prevent this attack?
12. Illustrate the process involved in understanding the inherent risk profile of a business with relevant tables and assumptions

Syllabus INFORMATION TECHNOLOGY-IT1

Module I: (8 Hours)

Introduction: CIA Triads, Hacking Impact, The Hacker Framework: Planning the test, Sound operations, Reconnaissance, Enumeration, Vulnerability Analysis, Exploitation, Final Analysis, and Deliverable. Integration Information Security Models: Computer Security, Network Security, Service Security, Application Security. Security Architecture Information Security Program: The Process of Information Security, Component Parts of Information Security Program, Risk Analysis, and Ethical Hacking.

Module II: (8 Hours)

The Business Perspective: Business Objectives, Security Policy, Previous Test Result. Business Challenges Planning for a Controlled Attack: Inherent Limitations Imposed Limitations, Timing Everything, Attack Type, Source Point, Required Knowledge, Multi-Phased Attack, Teaming and Attack Structure, Engagement Planner, The Right Security Consultant, The Tester, Logistics, Intermediates, Law Enforcement.

Module III: (8 Hours)

Preparing for a Hack: Technical Preparation, Managing the Engagement, Reconnaissance: Social Engineering, Physical Security, and Internet Reconnaissance.

Module IV: (8 Hours)

Enumeration: Enumeration Techniques, Soft Objective, Looking Around or Attack, Elements of Enumeration. Preparing for the Next Phase Exploitation: Intuitive Testing, Evasion, Threads and Groups, Operating Systems, Password Crackers, Rootkits, Applications, Wardialing, Network, Services, and Areas of Concern

Module V: (8 Hours)

Deliverable: The Deliverable, The Document, Overall Structure, Aligning Findings. Presentation Integration: Integrating the Results, Integration Summary, Mitigation, Defense Planning, Incident Management, Security Policy

Reference Books

1. Hacking: The Art of Exploitation Jon Erikson 2nd edition
2. The Basics of Hacking and Penetration Testing Patrick Engebretson 2nd edition
3. The Hacker Playbook 2 Peter Kim 1st edition
4. Penetration Testing Georgia Weldman 1st edition.

5. The Web Application Hacker's Handbook Dafydd Stuttard 2nd edition.
6. Stuart McClure, Joel Scambray and George Invernizzi, "Information Technology Security Secrets & Solutions", Tata McGraw-Hill Publishers, 2010 (6th Edition)
7. Ben Smith and Brian Komer, "Microsoft Windows Security Resource Kit", Prentice Hall of India, 2010

Syllabus and Course Plan (For 3 credit courses, the content can be for 40 hrs, and for 2 credit courses, the content can be for 26 hrs. The audit course in the third semester can have content for 30 hours).

No	Topic	No. of Lectures (40 Hours)
1	Module 1	8 Hours
1.1	Introduction: CIA Triads, Hacking Impact.	2
1.2	The Hacker Framework: Planning the test, Sound Operations, Reconnaissance, Enumeration, Vulnerability Analysis, Exploitation, Final Analysis, and Deliverable.	2
1.3	Integration Information Security Models: Computer Security, Network Security, Service Security, Application Security	2
1.4	Security Architecture Information Security Program: The Process of Information Security, Component Parts of Information Security Program, Risk Analysis, and Ethical Hacking	2
2	Module 2	9 Hours
2.1	The Business Perspective: Business Objectives, Security Policy, Previous Test Result	2
2.2	Business Challenges Planning for a Controlled Attack: Inherent Limitations Imposed Limitations,	2
2.3	Timing Everything, Attack Type, Source Point, Required Knowledge, Multi-Phased Attack	1
2.4	Teaming and Attack Structure, Engagement Planner,	2
2.5	The Right Security Consultant, The Tester, Logistics, Intermediates, Law Enforcement.	2
3	Module 3	7 Hours
3.1	Preparing for a Hack: Technical Preparation,	1
3.2	Managing the Engagement	2
3.3	Reconnaissance: Social Engineering	2
3.4	Physical Security, and Internet Reconnaissance.	2
4	Module 4	8 Hours
4.1	Enumeration: Enumeration Techniques, Soft Objective,	1

4.2	LookingAround or Attack, Elements of Enumeration.	2
4.3	Preparing for the Next Phase Exploitation: Intuitive Testing, Evasion, Threads and Groups, Operating Systems, INFORMATION TECHNOLOGY-IT	2
4.4	Password Crackers, Rootkits,Applications	2
4.5	Wardialing, Network, Services and Areas of Concern.	1
5	Module 5	8 Hours
5.1	Deliverable: The Deliverable, The Document, Overall Structure, Aligning Findings	2
5.2	Presentation Integration: Integrating the Results, Integration Summary	2
5.3	Mitigation, Defense Planning	2
5.4	Incident Management, Security Policy	2

